

Starting from the broad research topic “ISAC for UAVs,” the proposed framework retrieves and analyzes 434 papers and preprints spanning 2014–2026 and generates 70 candidate research ideas. Each idea is scored on a continuous 0–10 scale using eight weighted criteria: problem significance (20%), novelty defensibility (15%), physical-layer depth (20%), model closure (15%), practical relevance (10%), literature grounding (10%), scope discipline (5%), and execution feasibility (5%).

Rank	Score	Research Topic
1	9.86	Near-Field ISAC for Six-Dimensional UAV Pose Estimation During Landing
2	9.82	High-Speed UAV Sensing With 5G NR Signals: Ambiguity Analysis and Pilot Design
3	9.79	Quickest UAV Detection With Adaptive ISAC Beam Sweeping
4	9.76	Recovering UAV Micro-Doppler From Punctured OFDM Signals
5	9.73	Noncoherent Cell-Free ISAC for UAV Detection
6	9.71	UAV Uplink Signals for Passive Bistatic Airspace Sensing
7	9.69	Soft-Information-Aided UAV Detection With OFDM Signals
8	9.66	Numerology Selection for High-Mobility UAV ISAC
9	9.64	Coherent UAV Detection Under Range-Doppler Migration
10	9.61	Sensing-Assisted Beam Alignment for UAV Relaying With Information Causality
11	9.58	Ghost-Target Suppression in Multi-TRP ISAC for Multi-UAV Surveillance
12	9.55	Doppler-Resilient PRS Design for UAV Localization
13	9.53	Phase-Noise-Limited OFDM Sensing of High-Speed UAVs
14	9.50	UAV Data Collection With Joint Device Localization and Clock Calibration
15	9.47	Coherent Processing Interval Design for UAV-Borne ISAC Under Attitude Jitter
16	9.45	Three-Dimensional UAV Localization With Asynchronous Multi-TRP ISAC
17	9.42	Separating Doppler and Carrier Offset in Reciprocal UAV ISAC
18	9.40	Sensing-Aided Beam Tracking Under Fast UAV Attitude Variations
19	9.38	Measurement Scheduling for Cell-Free UAV Tracking
20	9.36	UAV-Enabled Backscatter ISAC for Batteryless Tag Localization
21	9.34	Multi-Band ISAC for Unambiguous UAV Tracking
22	9.32	Discriminative ISAC Waveform Design for UAV-Bird Recognition
23	9.30	Rate-Constrained Sensing-Statistic Feedback for UAV Target Tracking
24	9.28	Multi-TRP UAV Detection With Unknown Bistatic Reflectivities
25	9.26	Distributed Micro-Doppler Fusion for UAV Classification
26	9.24	Over-the-Air Phase Calibration for Cooperative UAV Sensing
27	9.22	Near-Field Beam Tracking for Low-Altitude UAV Communications
28	9.20	OTFS Sensing of Accelerating UAVs
29	9.18	Echo-Assisted Physical-Layer Authentication for Cellular UAVs
30	9.16	Sensing-Aided Grant-Free Access for Cellular-Connected UAVs
31	9.14	Finite-Blocklength ISAC for Timely UAV Detection
32	9.12	Sensing-Assisted Beam Switching Under Air-to-Ground Blockage
33	9.10	Sensing-Aided Robust Secrecy for UAV Communications With a Mobile Eavesdropper
34	9.08	Joint Trajectory and Waveform Design for UAV-Borne Target Tracking
35	9.06	Rotor-Speed Estimation From Cellular ISAC Echoes
36	9.04	Two-Way ISAC for Clock-Offset-Invariant UAV Swarm Localization
37	9.02	UAV-Borne Passive ISAC With Cellular Downlink Signals
38	9.00	UAV Illumination for Bistatic ISAC: Geometry and Power Allocation
39	8.98	Wideband Beam Tracking for UAV ISAC Under Beam Squint
40	8.96	Superimposed Pilots for UAV Sensing and Communications
41	8.94	Sensing With HARQ Retransmissions for Cellular-Connected UAVs
42	8.92	Active Sensing for UAV Jitter Tracking
43	8.90	UAV Detection With Unknown Noise Power in OFDM ISAC
44	8.88	Bistatic Diversity for UAV Detection Under Swerling Fading
45	8.86	Inter-Cell Coordination for UAV Detection in ISAC Networks
46	8.84	Task-Oriented Fronthaul Quantization for Cell-Free UAV Localization
47	8.82	One-Bit Fronthaul for Cell-Free UAV Detection
48	8.80	AFDM Sensing of UAVs With Doppler Spread
49	8.78	Low-Resolution ADCs for Bistatic UAV Sensing
50	8.76	Full-Duplex UAV ISAC With Residual Self-Interference
51	8.74	Constant-Envelope Waveform Design for UAV ISAC
52	8.72	Bistatic ISAC With a UAV Relay Under Direct-Link Interference

Rank	Score	Research Topic
53	8.70	Mobility-Assisted Airspace Sensing in Cellular UAV Networks
54	8.68	Sensing-Dwell Placement for UAV-Borne ISAC With Moving Targets
55	8.66	Synchronization-Aware Role Assignment for Multistatic UAV Sensing
56	8.64	Transmit-Receive Mode Selection for Multi-UAV Bistatic ISAC
57	8.62	Coherent UAV-Swarm ISAC With Imperfect Relative Positioning
58	8.60	Event-Triggered Sensing for UAV Beam Tracking
59	8.58	Robust UAV Communications With Sensing-Dependent Position Errors
60	8.56	Sparse UAV Formation Design for Distributed ISAC
61	8.54	Orientation-Robust Beamforming With Conformal UAV Arrays
62	8.52	Covert UAV Communications Under Public ISAC Signaling
63	8.50	Double-Pass RIS Beamforming for UAV-Borne ISAC
64	8.48	Doppler-Aware Aerial RIS for Terrestrial ISAC
65	8.46	User-Centric Clustering for Cellular UAV Communication and Localization
66	8.44	Information-Causal ISAC for UAV Base Stations With Wireless Backhaul
67	8.42	True-Time-Delay Beamforming for Wideband UAV ISAC
68	8.40	Pointing-Robust FSO Backhaul for UAV-Borne ISAC
69	8.38	Adaptive Beam Probing for UAV ISAC Under Markov Blockage
70	8.20	Cache-Aware UAV Multicast ISAC With Sensing-Freshness Constraints